
Current
Electricity

Current Electricity

CURRENT

Current is defined as rate of flow of charge.

AVERAGE CURRENT

Charge flown per unit time where the time interval is long.

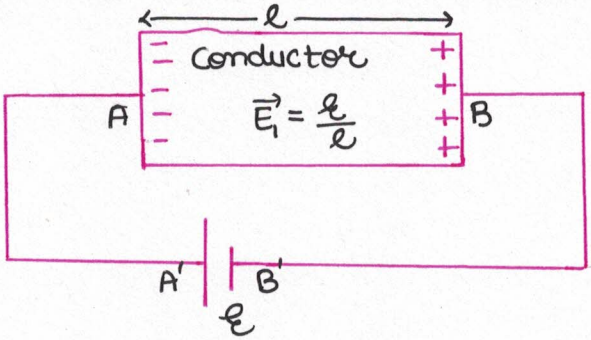
$$I_{avg} = \frac{Q}{\Delta t}$$

INSTANTANEOUS CURRENT

Charge flown per unit time where the time interval is very small.

$$I_{ins.} = \frac{dQ}{dt}$$

ANALYSIS FOR CONDUCTOR NOT IN ELECTROSTATIC CONDITION



$$V_A = V_A' \quad \text{and} \quad V_B = V_B'$$

The potential difference in wire is zero. Still the charge is flowing in it. Its because they are resistance less wire.

$$V = IR$$

(V → zero, R → zero)

Now, $V_A - V_B = \mathcal{E}$ creates potential difference and it creates electric field $= \frac{\mathcal{E}}{\ell} = E$.

but this case is a different from that, one discussed earlier. Here the charge won't stop after coming to A and B. They will go into the wires. Hence the charge keeps on moving inside the conductor & the field won't be zero.

DRIFT VELOCITY OR AVERAGE VELOCITY

e^- would not flow in a straight line under constant acceleration. Instead they would flow in a zig-zag motion because of the collision of the e^-.

Let u_i be the velocity of an e^- just after the collision and let it moves for t_i time after the collision.

$$\vec{v_i} = \vec{u_i} - \frac{e\vec{E}}{m} t_i$$

$$\text{avg.}\vec{v} = \frac{\sum \vec{v_i}}{N} = \frac{\sum \vec{u_i} - \frac{e\vec{E}}{m}\sum t_i}{N}$$

$\vec{u}$ is a random vector

$$\therefore \sum \vec{u_i} = 0$$

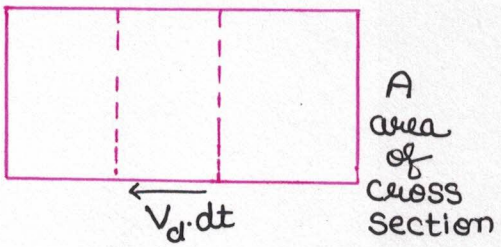
$$\text{avg.}\vec{v} = \frac{-e\vec{E}}{m}\left(\frac{\sum t_i}{N}\right)$$

$$\boxed{\text{avg.}\vec{v} = v_{drift} = \frac{-e\vec{E}}{m}(\tau)}$$

- τ is relaxation time i.e. the time interval between the collision of two e^-.
- τ depends on nature of material and temperature.

n: no. of e^- per unit volume

A: area of cross section



Volume of section being considered $= A v_d . dt$

$$\text{Current} = \frac{\text{charge flown}}{t} = \frac{enA v_d . dt}{dt}$$

$$\boxed{I = enA v_d}$$

CURRENT DENSITY ($\vec{J}$)

Current density is defined as charge flowing per unit area. Its direction is opposite to that of drift velocity or is same as that of electric field.

$$\vec{J} = -ne\vec{v_d}$$

$$\vec{J} = \left(\frac{ne^2\tau}{m}\right)\vec{E} = \sigma\vec{E}$$

$$\vec{J} = \sigma\vec{E}$$

(σ is the conductivity of the material)

Now current can be defined as

$$I = \int \vec{J}.d\vec{s}$$

(on increasing temp, τ decreases, σ decreases,)
but its only for metal

$$\therefore \sigma \propto \frac{1}{\text{temp.}}$$

CONDUCTIVITY

Resistivity is defined as the reciprocal of conductivity.

$$\rho = \frac{1}{\sigma}$$

RESISTANCE

$$\vec{J} = \sigma\vec{E}$$

$$\frac{I}{A} = \frac{1}{\rho}\vec{E}$$

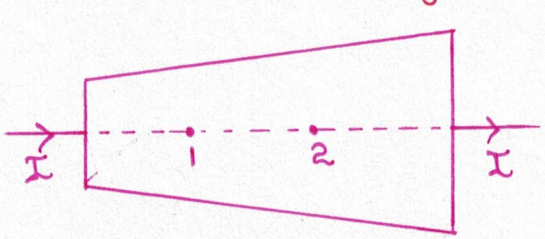
$$\frac{I}{A} = \frac{1}{\rho}\left(\frac{V}{\ell}\right)$$

$$\frac{V}{I} = \frac{\rho\ell}{A} = \text{Constant} = \text{Resistance}(R)$$

$$\frac{V}{I} = R$$

Resistance is a property of a substance to oppose the flow of charge.

Que.) I is constant
(a) Find where the field is more.
(b) Find where the drift velocity is more.



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I = Constant

$\therefore$ $J.ds$ = Constant

at (1) field is more

and at (1) drift velocity is more.

$$I = \sum n_i v_{di} A_i \ell_i$$

ℓ_i → charge of each particle

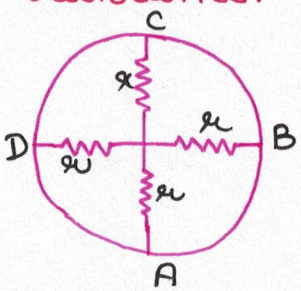
A_i → area of cross section

n_i → charges per unit volume

v_{di} → drift velocity

(It can be used if particles with different charges are flowing from areas of different cross-section)

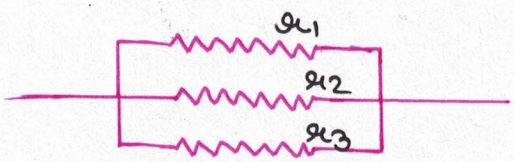
Que.) Find eq. resistance.



$R_{eq} = \frac{R}{4}$

PARALLEL ARRANGEMENT

The arrangement in which one end of the different resistors are connected to one common point and other end to another common point.



$$I = I_1 + I_2 + I_3 + \text{------}.$$

$$\frac{V}{R_{eq}} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + \text{-----}$$

$$\boxed{\frac{1}{R_{eq.}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

$$\frac{1}{R_{eq.}} \leqslant \frac{1}{R_i}$$

SERIES ARRANGEMENT

The arrangement in which the resistors are joined end to end or we can say they are connected one after the other.



$\therefore$ Here $V_i \propto R_i$

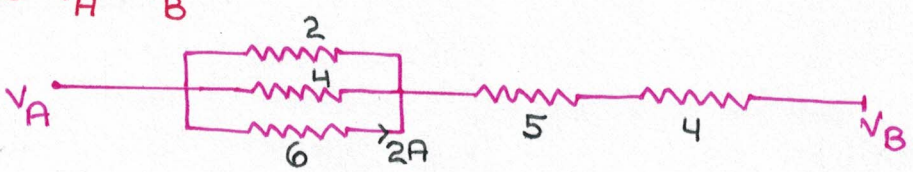
$$V = V_1 + V_2 + V_3 + V_4 + \text{-------}$$

$$IR_{eq.} = IR_1 + IR_2 + IR_3 + IR_4 + \text{------}$$

$$\boxed{R_{eq} = R_1 + R_2 + R_3 + \text{----}}$$

$$R_{eq} \geq R_1$$

Que.) Find $V_A - V_B$



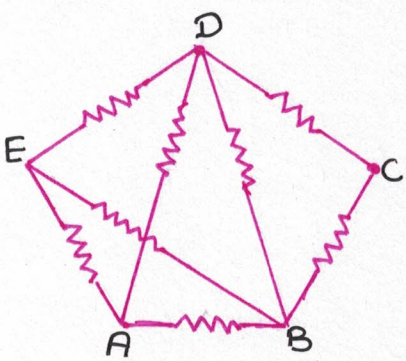
$$V_A - V_B = (-111)\,V$$

Que.) There are 1002 points. Each point is connected to other with resistance R. Calculate the equivalent between two of them.

As the point E is, as it is D and as it is C. E is connected to all, D is also connected and C too.

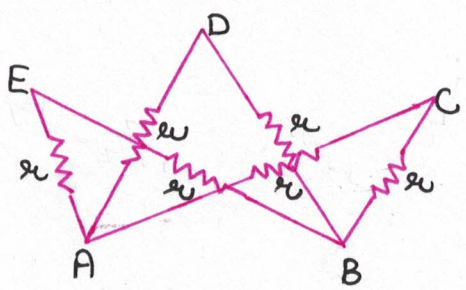
$\therefore$ we can say that E, C, D are equipotential points.

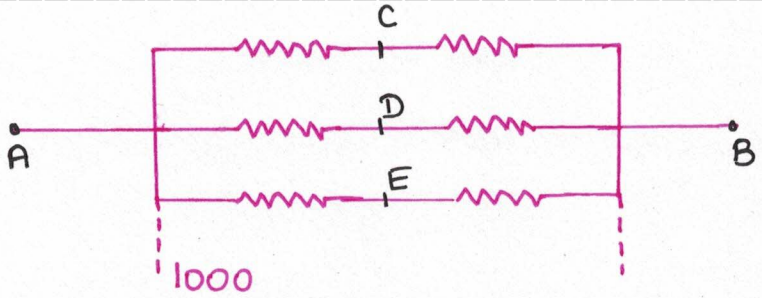
$$\therefore V_E = V_C = V_D$$



So any resistor connected to any of these points E, C & D can be removed.

$\therefore$ The above circuit is





$$\frac{1}{Req.} = \frac{3}{2r} + \frac{1}{r}$$

$$= \frac{5-2}{2r} + \frac{1}{r} = \frac{n-2}{2r} + \frac{1}{r}$$

In case of 1002 points

$$\frac{1}{Req} = \frac{n-2}{2r} + \frac{1}{r}$$

$$= \frac{1000}{2r} + \frac{1}{r}$$

$$= \frac{500}{r} + \frac{1}{r}$$

$$Req. = \frac{r}{501}$$

KIRCHOFF'S LAWS

1. JUNCTION LAW

Sum of incoming and outgoing currents at the junction are equal.

OR

If incoming current are considered positive and outgoing current are considered negative then their algebraic sum is zero.

(It is followed from conservation of charge)

KIRCHOFF'S VOLTAGE LAW / LOOP LAW

(It is followed from conservation of energy)

If a charged element crosses through the same point after moving through a loop then net work done on this charge is 0.

$$\therefore \Sigma (\Delta q, V_i) = 0$$

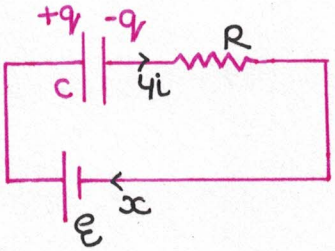
$$\Delta q, \Sigma V_i = 0$$

$$\boxed{\Sigma V_i = 0}$$

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Loop Law

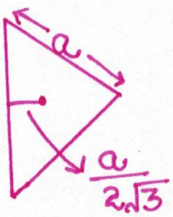
$$+\mathcal{E} - \frac{q}{C} - iR = 0$$

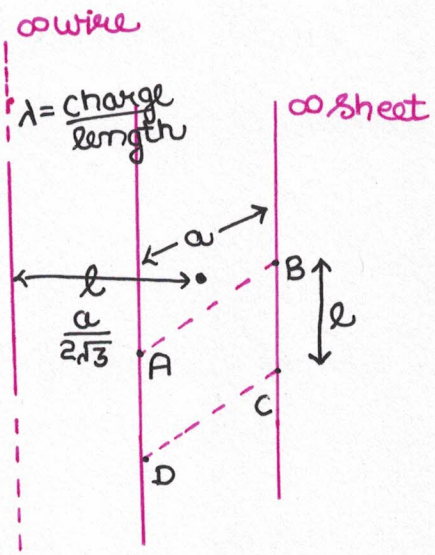


Half Loop Law

$$+\mathcal{E} - \frac{q}{C} = V_y - V_x = +iR$$

Que: Calculate flux through ABCD.





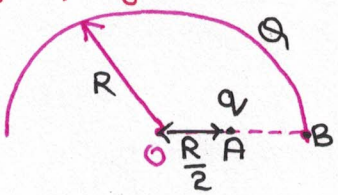
∴ flux through whole sheet
$= \frac{1}{3}^{rd}$ of total flow

∴ flux through the area being considered $= \frac{1}{3}\left(\frac{\lambda \ell}{\epsilon_0}\right)$

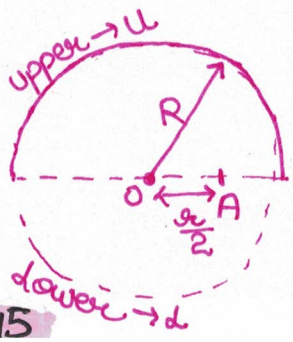
In Case $\ell \to 0/2$, then form a square.

In Case $\ell \to \sqrt{3}a/2$, then form a hexagon.

Que: A uniformly charged hemisphere is given. Find the work done in taking q from A to B.



Because of the hemisphere shell-U, let us say W is the work done. Then, W is also the work done in presence of hemi shell L. Thus in presence of both hemi-shells, work done is 2W.

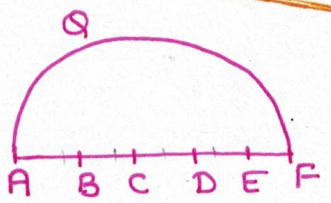


$\therefore 2W = 0$ (as potential at A & B are equal)

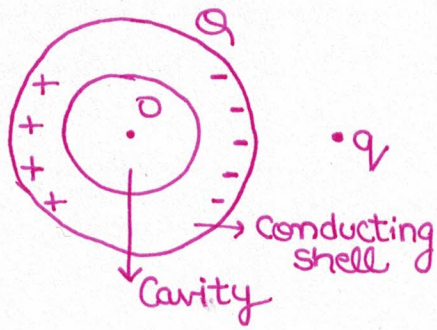
$\therefore W = 0$

Thus, we can say that all points on a hemispherical base are equipotential.

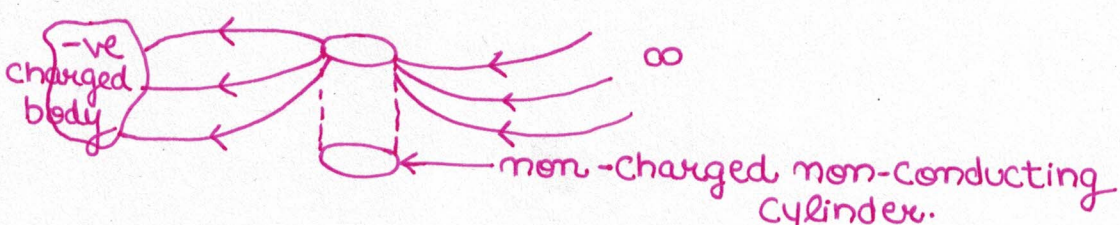
NOTE 1: All points on the base of a hemispherical shell are equipotential.
(Reason is in the previous que)



NOTE 2: Because of q, there is non-uniform distribution of Q.
But if we place 'q' inside the cavity, then '-q' will induce in the surface of cavity. Thus +q is induced on the outer surface of the shell but this charge will distribute uniformly so as to maintain the equipotentiality of the conducting shell.



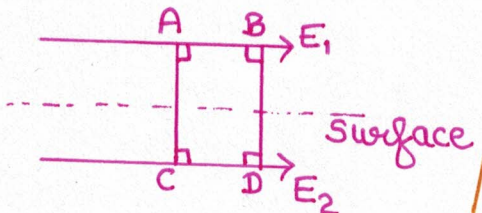
NOTE 3:



as $V_\infty = 0$ and electric field lines go from higher potential to lower potential, $V_{cylinder} < 0$. Thus potential of cylinder is negative.

NOTE 4: It is not possible that parallel electric field lines are existing with different magnitude.

As electrostatic force is a conservative force, $\int E.dl$ for closed loop $= 0$

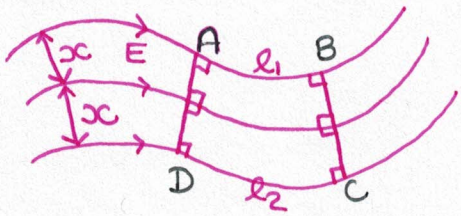


$\therefore E_1 l + E_2 l + 0 + 0 = 0$

$E_1 l$ → for AB, $E_2 l$ → for CD, 0 → for BD, 0 → for AC

It is not possible.

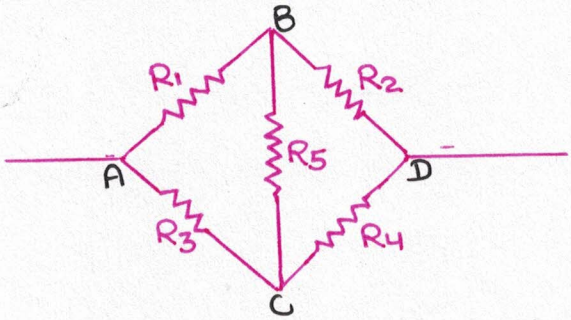
NOTE 5: Parallel electric field lines with same magnitude exist in wave shape.



$El_1 + El_2 + 0 + 0 = 0$

Not possible.

WHEATSTONE BRIDGE

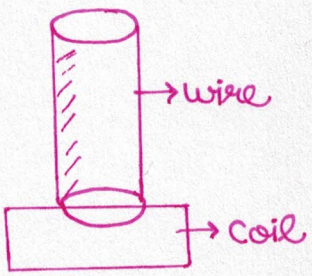


If $R_1 \times R_4 = R_3 \times R_2$

then R_5 is as good as non-existing.

GALVANOMETER

We use a wire (from which no current is passing) and it is joined to a coil at one end. Current passes through the coil and it creates a magnitic field (B). It creates a torque (τ) on the wire which tries to create torsional strain.



$\tau = nBIA$ → area of coil, magnetic field, current, no. of turns in coil

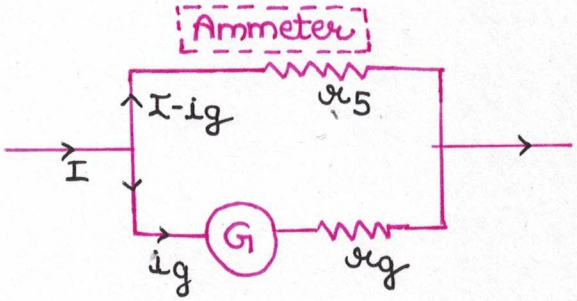
$\therefore \tau \propto I$

also, $\tau \propto \theta$ and $\tau = C\theta$ (torsional constant)

GALVANOMETER AS AMMETER

Galvanometer does not allow to pass a large amount of current through it.

So we make an arrangement so that only a small fraction of it (i_g) pass through it.



$r_s << r_g$ so that $I - i_g >>> i_g$

$(I - i_g) r_s = i_g \cdot r_g$ (voltage is constant)

$$I = i_g \left(1 + \frac{r_g}{r_s}\right)$$

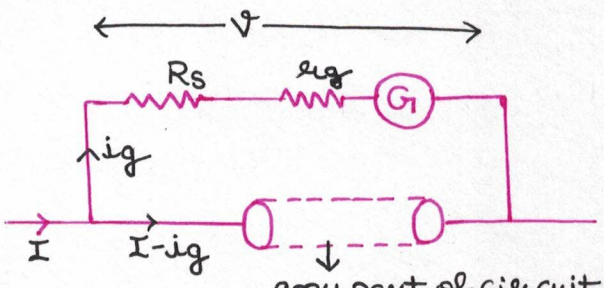
arrangement is called Ammeter.

GALVANOMETER AS VOLTMETER

$R_s >>> r_g$

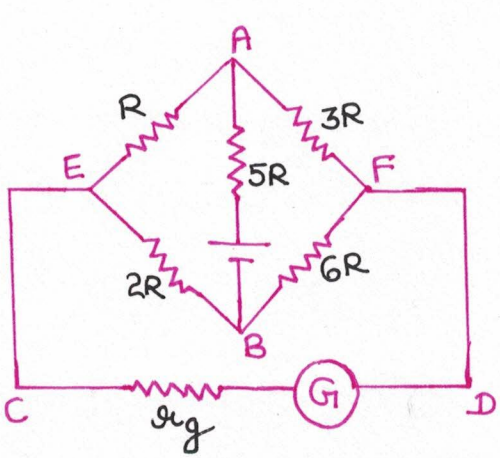
So that $I - i_g >>> i_g$

$V = i_g (R_s + r_g)$

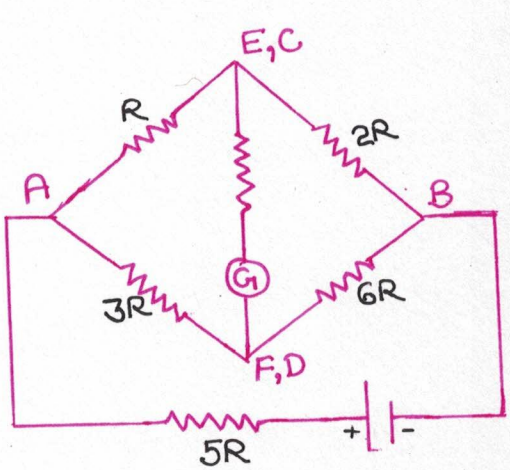


This arrangement is called voltmeter.

Important thing to be noticed



Reading of Galvanometer = 0



This is the same circuit shown before.

POWER OF A RESISTOR

Power $= IV$ (for every resistor)

Power $= \frac{V^2}{R} = I^2R$ (for ohmic resistors)
↓
which follows ohm's law.

$$P = \frac{W}{t} = \frac{H}{t} = \frac{V.q}{t} = VI$$

Que.) Heaters A & B respectively boil a bucket of water in 4 min & 6 min. In what time they will boil the bucket of water if used in series simultaneously.

$$P_1 = \frac{V^2}{R_1} \Rightarrow H_1 = \frac{V^2 t_1}{R_1} \qquad \text{Req.} = (R_1 + R_2)$$

$$P_2 = \frac{V^2}{R_2} \Rightarrow H_2 = \frac{V^2 t_2}{R_2}$$

$$H_1 = H_2 \Rightarrow \frac{V^2 t_1}{R_1} = \frac{V^2 t_2}{R_2} = H$$

we can say, $\frac{V^2 t_1 - V^2 t_2}{(R_1 + R_2)} = \frac{V^2 t_1}{R_1} = \frac{V^2 t_2}{R_2} = H$

(by compon. & Dividend)

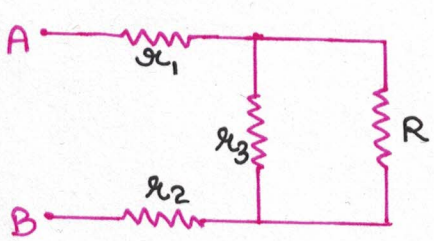
$$\frac{V^2(t_1 + t_2)}{(R_1 + R_2)} = H$$

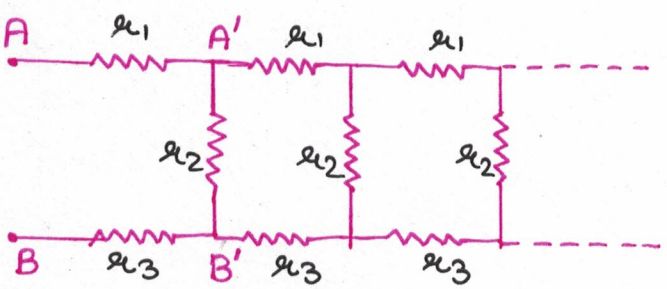
$\therefore t = t_1 + t_2 = 10\text{min.}$

INFINITE SERIES OF RESISTORS

Case 1:

an equivalent circuit can be drawn





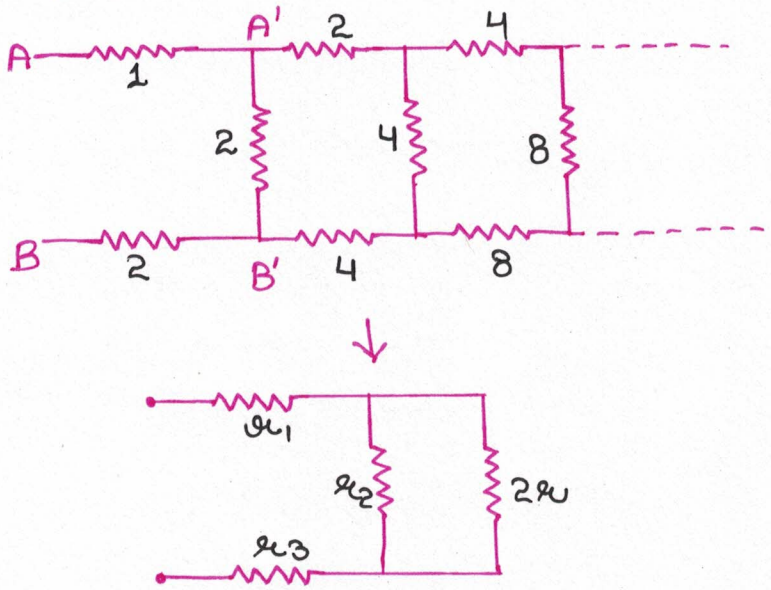
$$R = \left(\frac{R r_3}{R + r_3}\right) + r_1 + r_2$$

$$R^2 - R(r_1 + r_3) - (r_1 + r_2 + r_3 r_2) = 0$$

$$X^2 - R(r_1+r_3) - r_2(r_1+r_3) = 0$$

Solve for X.

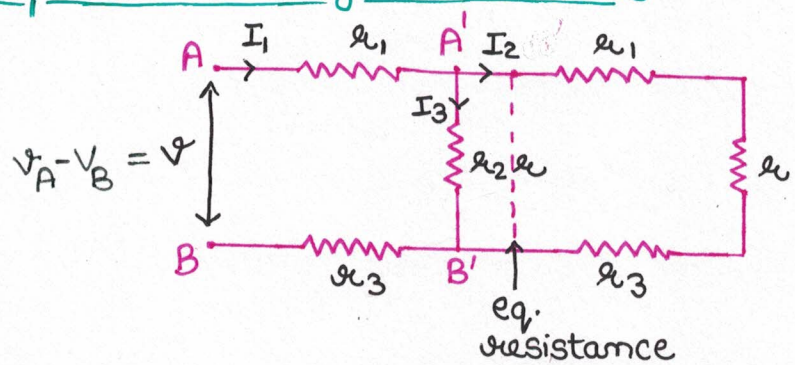
Case 2: If each resistance of a resistance network is multiplied by a common quantity, equivalent also multiplies with that.



$$r = r_1 + r_3 + \frac{Kr \cdot r_2}{r_2 + Kr}$$

In this case $K = 2$

Important thing in Case-1



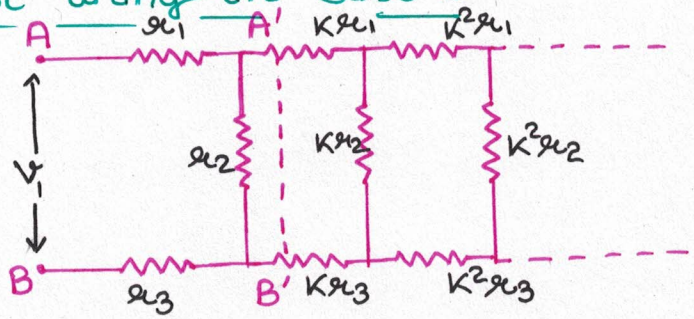
$$I_2 r = (I_1 - I_2) r_2$$

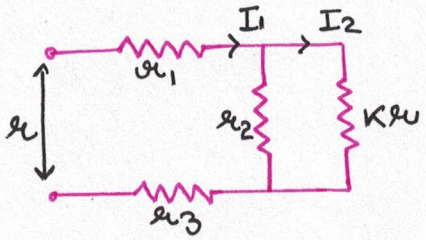
$$I_2 = \left(\frac{r_2}{r + r_2}\right) I_1$$

$$V_A' - V_B' = V\left(\frac{r_2}{r_2 + r}\right) \quad \left(\text{putting } I_1 = \frac{V}{r}\right)$$

We can always convert equipotential points into a single point.

Important thing in Case-2





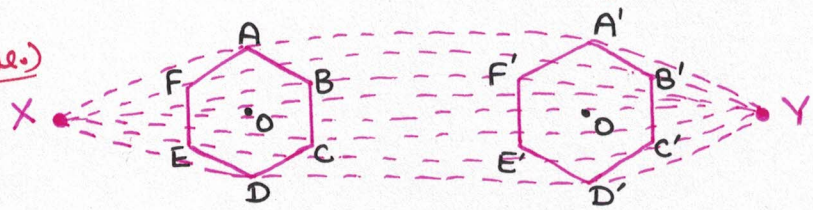
$$I_2 = \left(\frac{R_2}{R_2 + KR_2}\right) I$$

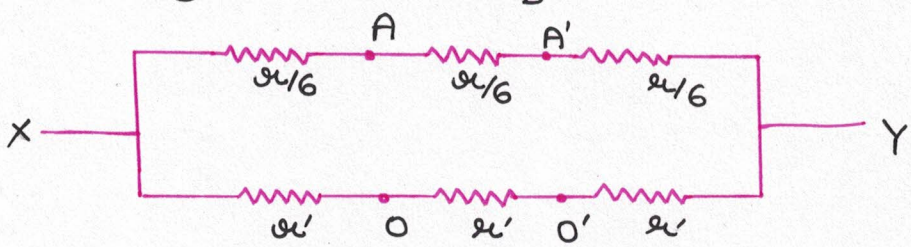
$$V_2 = KR \times I_2$$

$$V_2 = V_A' - V_B' = \left(\frac{KR_2}{R_2 + KR_2}\right) V_1$$

Questions based on symmetry:

Que.



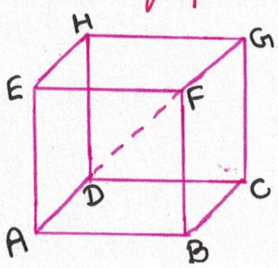


$$R_{eq.} = \frac{3R}{7}$$

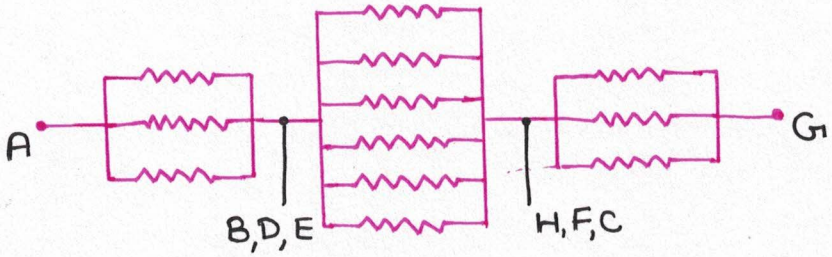
Que. Eq. resistance b/w the points representing body diagonal where each side represents resistance in a cube.

E, B, D are equipotential

& H, F, C are equipotential

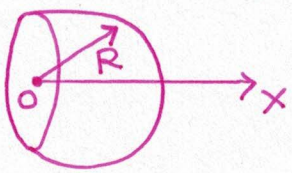


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$$R_{eq.} = \frac{5R}{6}$$

Que.) Find the current through the hemispherical surface.



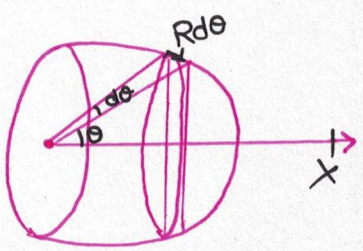
$$\vec{J} = J_0\left(\frac{x}{R}\right)\hat{i}$$

$$I = \int J.ds$$

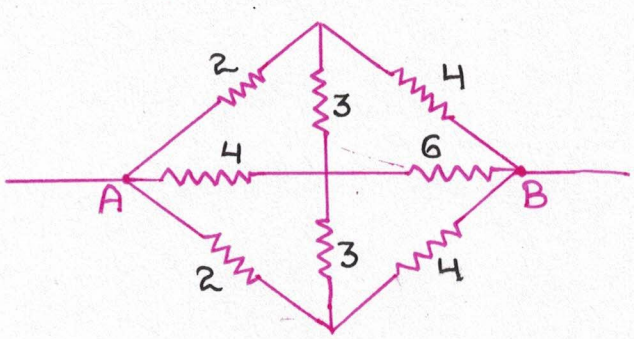
$$= \int 2\pi R^2 \sin\theta \cdot J_0 \cos^2\theta$$

$$= J_0\, 2\pi R^2 \int_0^R x^2.dx$$

$$I = \frac{2}{3} J_0 (\pi R^2)$$

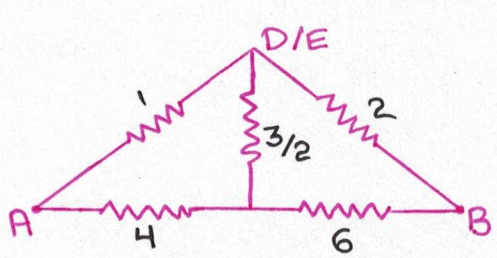


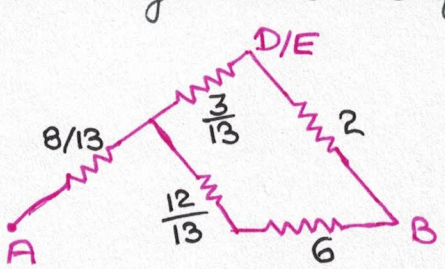
Que.) Find the equivalent resistance b/w the points A & B.



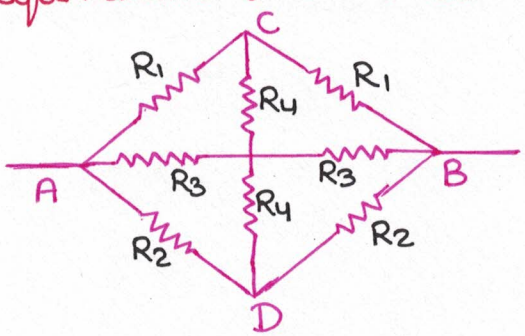
Input-output symmetry exists i.e. input-output points are interconvertible then take current same about some points.

In this case axial symmetry exists. So fold it.



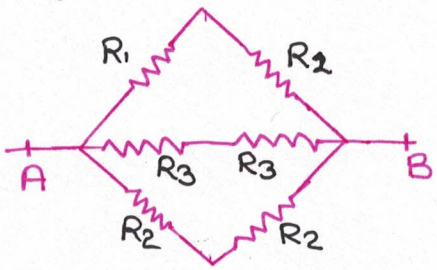


Que.) Find out equivalent resistance b/w the points A and B.



In this case axial symmetry does not exist but vertical symmetry exists ie A & B can interchange their positions and thus we can fold in a way so that A and B can overlap.

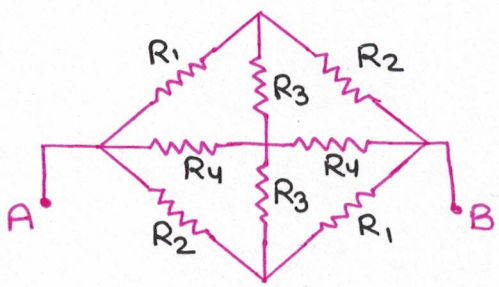
Thus current would be same from both R_1, both R_2, both R_3. That means no current is passing through Req.



(Simplified form)

(Rotational Symmetry:- If by rotating the circuit by some particular angle, the same circuit is formed then the current through the substituting parts is same.)

Que.)



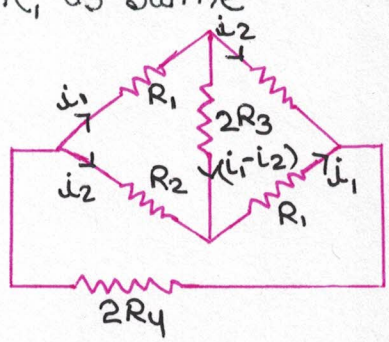
Rotate about O by 180°

Circuit remains same

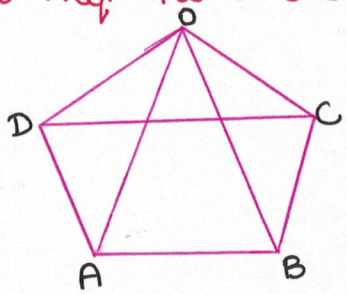
∴ we can say current from both R_1 is same

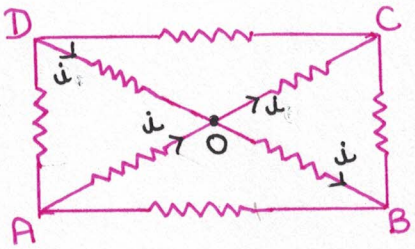
Similarly we can say for R_2.

Thus the circuit can be simplified as,

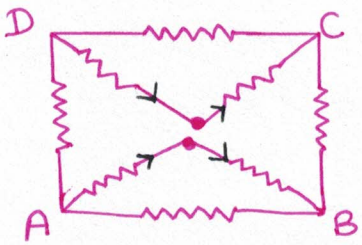


Que.) Find Req. b/w A & B. Each side has a resistance R.





(by rotational symmetry and input-output symmetry we can say O can be break into two points)

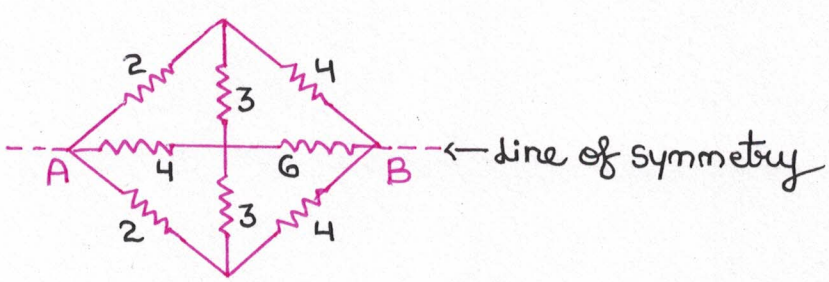


$$Req = \frac{8}{15} R$$

CONCLUSION OF SYMMETRY

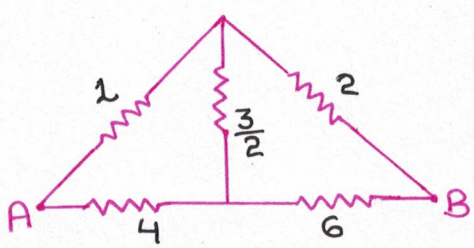
1. Axial Symmetry :- Symmetry about the line joining the points about which resistance is to be found.

Eg:



In this case fold the circuit about line of symmetry and multiply the resistances by 1/2 which are getting overlapped.

Simplified form ⇒



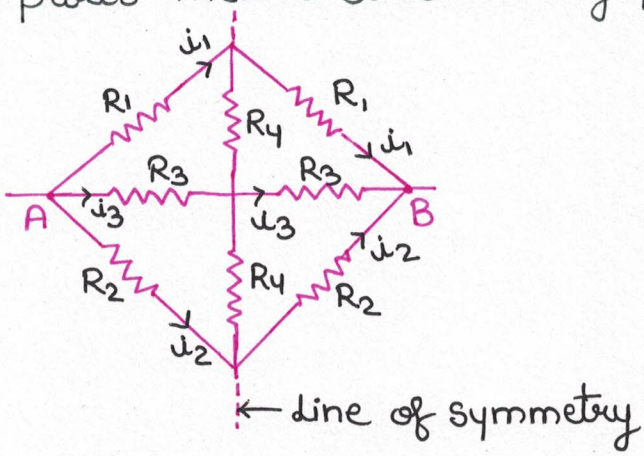
2.) Vertical Symmetry / Input-output symmetry :-

If there exists a line of symmetry and on folding about it, the points about which resistance is to be

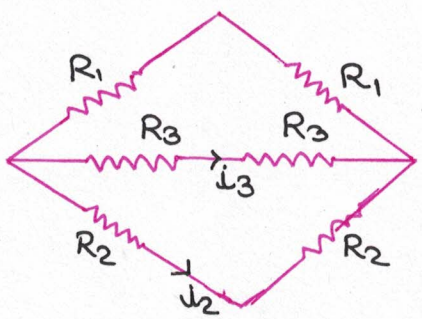
found gets overlap then current through the symmetrical parts of the circuit is same.

Symmetrical parts means substituting parts.

Eg: -

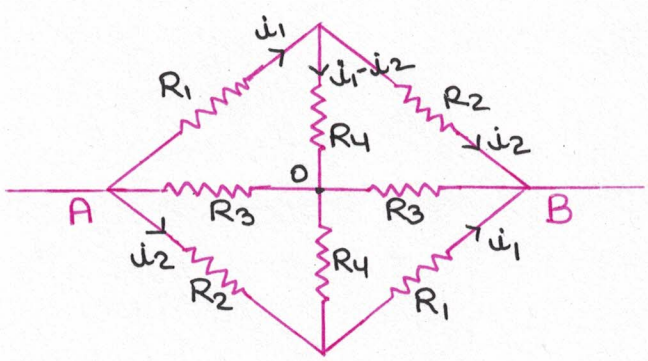


Simplified form -



3.) **Rotational Symmetry:** If by rotating some angle, the same circuit is formed as it was, then the current through the substituting parts is same.

Eg:

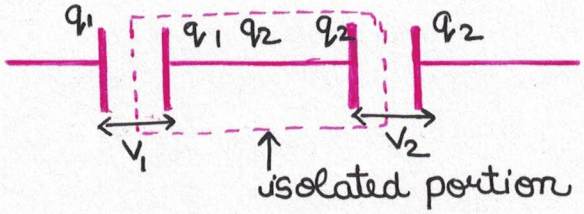


Rotate by 180° about O

∴ Rotational symmetry exists.

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SERIES COMBINATION OF CAPACITORS



$q_2 - q_1 = 0$

$q_2 = q_1$

$\therefore$ net charge should be 0.

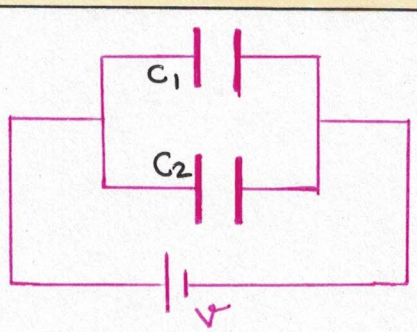
Thus we can say in series combination, charge remains same

$$V = V_1 + V_2$$

$$\frac{q}{C_{eq}} = \frac{q}{C_1} + \frac{q}{C_2}$$

$$\boxed{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}}$$

PARALLEL COMBINANATION OF CAPACITORS



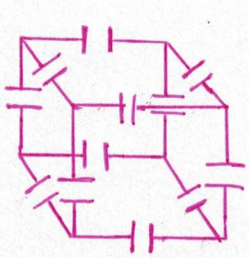
Voltage across all the capacitors is same

$$Q = Q_1 + Q_2$$

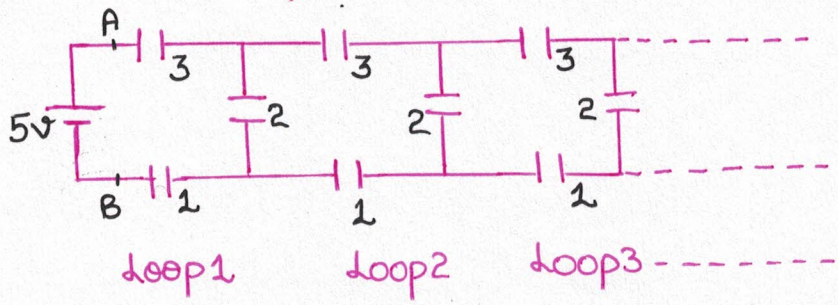
$$VC_{eq} = VC_1 + VC_2$$

$$\boxed{C_{eq.} = C_1 + C_2}$$

Que:) A cube with capacitors C on all sides is there. Find eq. Capacitance.



Que.) Find the charge on capacitor of 3μF of loop 10.



let $C_{eq} = x$

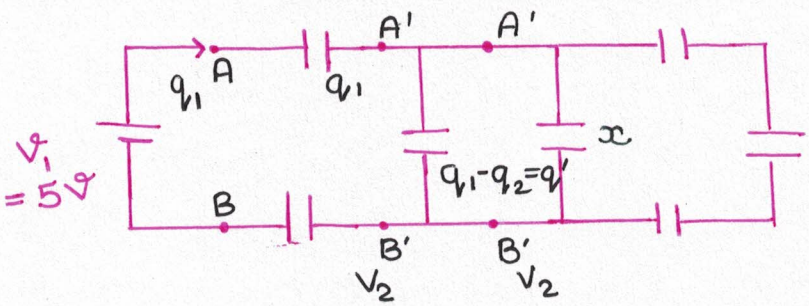
$$\frac{1}{x} = \frac{1}{1} + \frac{1}{2+x} + \frac{1}{3}$$

$$\frac{6+3x+3+2+x}{3(2+x)} = \frac{1}{x}$$

$$11x + 4x^2 = 6 + 3x$$

$$4x^2 + 8x - 6 = 0$$

$$2x^2 + 4x - 3 = 0$$



$$V_1 = \frac{q_1}{x}, \quad V_2 = \frac{q_2}{x} = \frac{q_1 - q_2}{x} = \frac{q'}{x}$$

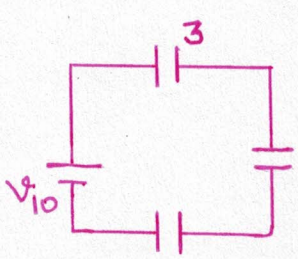
$$q_2 = q_1\left(\frac{2}{2+x}\right)$$

$$q' = q_1 - q_2 = q_1\left(\frac{x}{2+x}\right)$$

$$V_2 = V_1\left(\frac{x}{2+x}\right)$$

Similarly, by seeing the pattern,

$$V_{10} = V_1\left(\frac{x}{2+x}\right)^9$$



$$q_{req} = V_{10} \times C_{eq}$$

$$= V_1 \times \left(\frac{x}{2+x}\right)^9 \times x$$

$$= V_1\left(\frac{x}{2+x}\right)^9 x$$

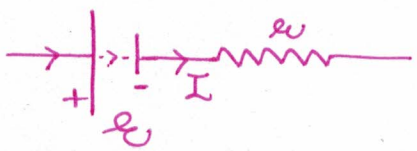
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ARRANGEMENT OF BATTERIES

- 2 ideal batteries with different e.m.f. can not be arranged in parallel.
- It is possible to increase the potential difference of a battery of emf in two ways:

(a) If current flows in opposite to the usual sense.

$$V = \mathcal{E} + Ir$$



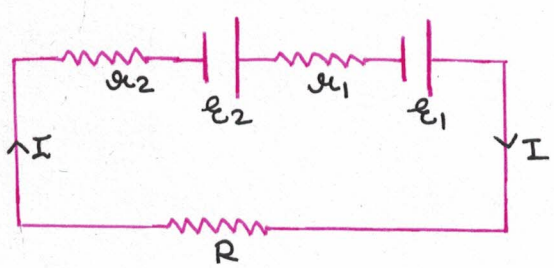
(b) If current flows in its usual sense but in very large amount.

$$V = \mathcal{E} - Ir$$

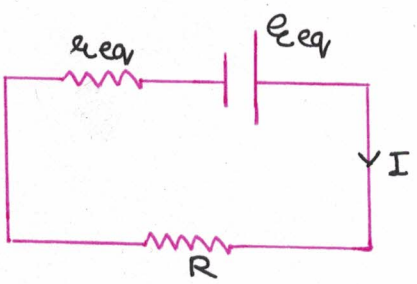
and $Ir > 2\mathcal{E}$



BATTERIES IN SERIES



$\Rightarrow$



$$I = \frac{\mathcal{E}_1 + \mathcal{E}_2}{R + (r_1 + r_2)}$$

$$I = \frac{\mathcal{E}_{eq.}}{R + r_{eq.}}$$

$$\mathcal{E}_{eq} = \mathcal{E}_1 + \mathcal{E}_2 + \text{-----} = \sum \mathcal{E}_i$$

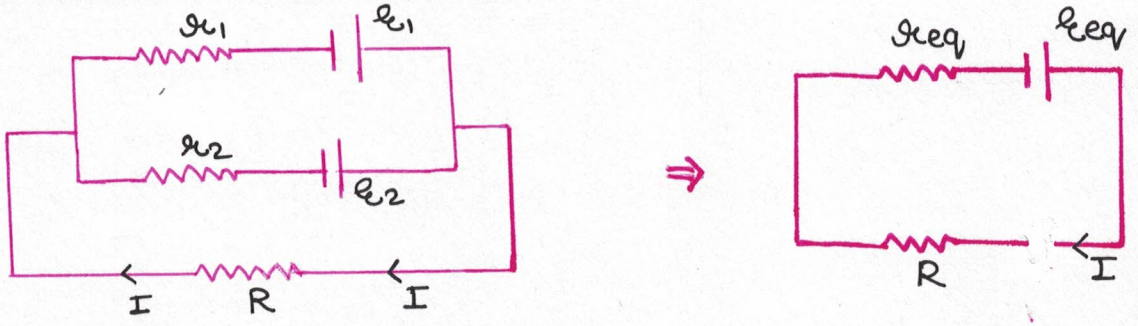
$$r_{eq} = r_1 + r_2 + \text{---} = \sum r_i$$

Que.) 12 batteries each of emf 2V and internal resistance of 0.5 Ω are connected in series and joined to a resistance of 6 Ω. 2 batteries by mistake are joined with opposite polarity. Calculate the potential difference b/w their terminals.

$$I = \frac{8 \times 2}{6 + 12 \times \frac{1}{2}} = \frac{4}{3}$$

$$V = \mathcal{E} + Ir = 2 + \frac{4}{3} \times \frac{1}{2} = \frac{8}{3} V$$

BATTERIES IN PARALLEL



by loop laws :

$$\mathcal{E}_1 - IR - i r_1 = 0$$

$$i_1 = \frac{\mathcal{E}_1}{r_1} - \frac{IR}{r_1}$$

$$I = \sum \frac{\mathcal{E}_i}{r_i} - IR \sum \frac{1}{r_i}$$

$$I = \frac{\sum \mathcal{E}_i / r_i}{1 + R \sum 1/r_i} = \frac{\sum \frac{\mathcal{E}_i}{r_i} / \sum 1/r_i}{\frac{1}{\sum 1/r_i} + R}$$

$$\therefore \mathcal{E}_{eq} = \frac{\sum \mathcal{E}_i / r_i}{\sum 1/r_i}$$

$$r_{eq} = \frac{1}{\sum r_i}$$

SHORT - CIRCUITING

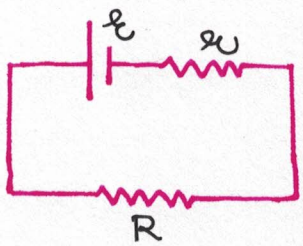
When the terminals of a circuit element are joined through a resistanceless wire, that element is called short circuited. Combinely, they behave as zero resistance for the circuit.

MAXIMUM POWER TRANSFER THEOREM

A battery of emf $\mathcal{E}$ and internal resistance r is given. We have to find R for which power delivered is maximum.

$$I = \frac{\mathcal{E}}{R + r}$$

$$P_{delivered} = I^2 R$$



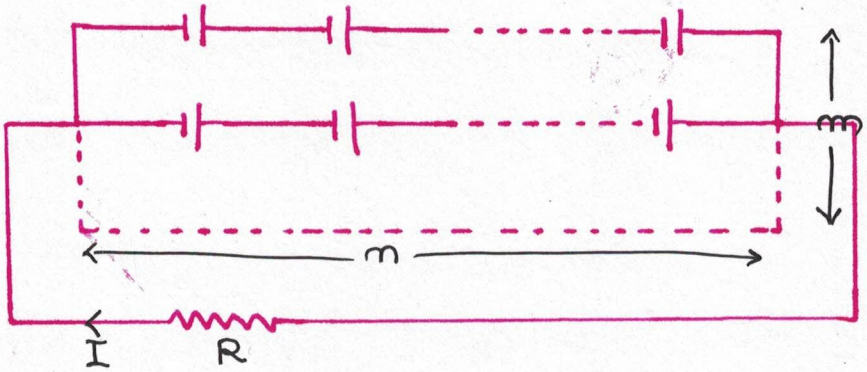
$$P = \frac{\mathcal{E}^2 R}{(R+r)^2} = \frac{\mathcal{E}^2}{\frac{(R-r)^2}{R} + 4r}$$

For P to be max.,

$$\frac{(R-r)^2}{R} = 0$$

$$\therefore \boxed{R = r}$$

Que.) Find the ratio of m/n for which I is max.

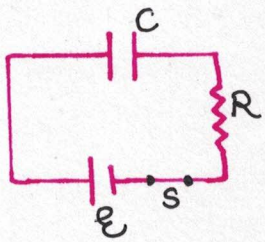


$$I = \frac{n\mathcal{E}}{\frac{nr}{m} + R} = \frac{nm\mathcal{E}}{(\sqrt{nr} - \sqrt{mr})^2 + 2\sqrt{nmRr}}$$

For I to be max., $nr = mR$

R-C CIRCUIT (ONLY ANALYSIS)

- As soon as the switch is switched on, the current that will flow in the circuit will have a definate value because if current becomes infinite then potential difference across resistance R becomes infinite. But if resistance would not be there, then infinite current flows but for very small time.



- If there is a definate current through a capacitor charge on it, do not change immediately.

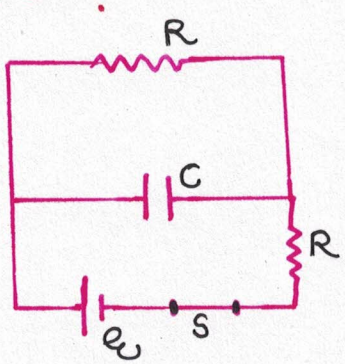
$$\Delta q = \int_0^{\Delta t} I \cdot dt \to 0$$

If $I = K$ and $\Delta t \to 0$

- In a circuit involving capacitor, a steady state means the charge on the capacitor has become constant. It means current through the capacitor becomes zero.

- Current immediately after the switch is switched on is $I = \mathcal{E}/R$, but after long time it will become zero.

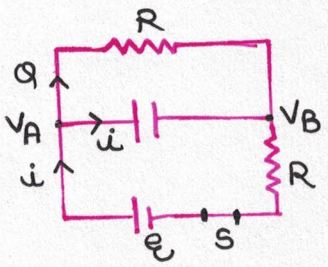
Que.) (a) what is the current, immediately after the switch is switched on?
(b) what is the current in the steady state?
(c) what is the charge on the capacitor initially and in steady state?



(i) $V_A - V_B = 0$ (initially)

$\therefore I = \frac{\mathcal{E}}{R}$

INITIAL

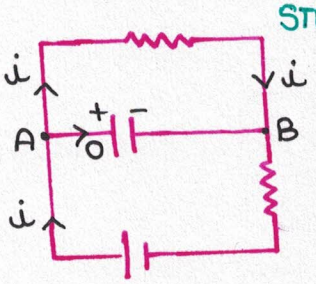


(ii) $I = \frac{\mathcal{E}}{2R}$

$V - iR = 0$ (by loop law)

$V = \frac{\mathcal{E}}{2R} \times R = \frac{\mathcal{E}}{2}$

STEADY STATE



(iii) $Q_{final} = CV$

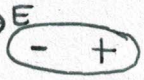
$Q_{final} = \frac{C\mathcal{E}}{2}$

$Q_{initial} = 0$

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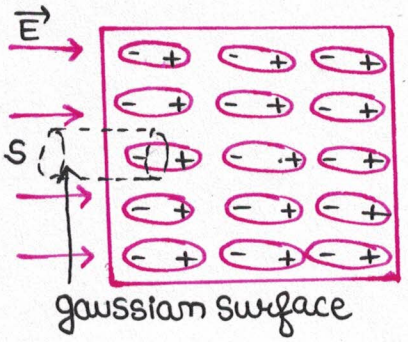
PROPERTIES OF DIELECTRIC

- We consider only those cases in which field is always normal to the surface of dielectric.
- When ext. field is applied, then if

(a) molecules are neutral then they will become like  and will behave as dipole.

(b) molecules have their dipole, then initially their dipoles were randomly directed but after applying the field they will become uniformly directed.

$$\text{Dielectric Constant} = \frac{E}{E_{ind}}$$



By gauss law,

$$-ES + E_{ind}S = \frac{\sigma S}{\epsilon_0}$$

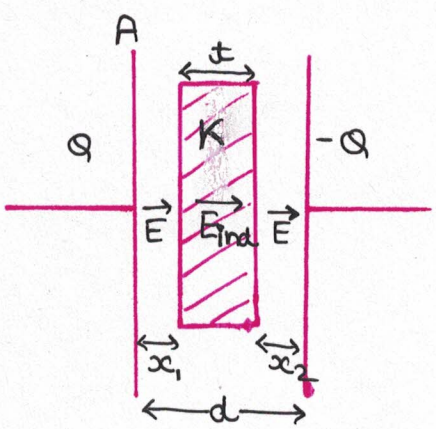
$$\sigma = (E_{ind} - E)\epsilon_0$$ (ϵ_0 is -ve charge surface density)

- Equivalently it appears that negative & positive charges are created at the surface of the dielectric. These charges are known as bound charges.
- When electric field is applied on surface of dielectric normally, a dipole moment / unit volume is created in the dielectric reducing the net field inside dielectric.

STANDARD RESULTS FOR DIELECTRIC

Case 1:

We know, how to solve it by thinking it in series and introducing an imaginary plate.



lets solve it fundamentally,

$$\Delta V = Ex_1 + Ex_2 + E_{ind}t$$

$$\Delta V = \frac{\sigma}{\epsilon_0}(x_1 + x_2) + \frac{\sigma}{K\epsilon_0}t$$

$$\Delta V = \frac{\sigma}{\epsilon_0}(d-t) + \frac{\sigma}{K\epsilon_0} t$$

$$\Delta V = \frac{Q}{A\epsilon_0}(d-t) + \frac{Q}{AK\epsilon_0} t$$

$$\frac{Q}{\Delta V} = C$$

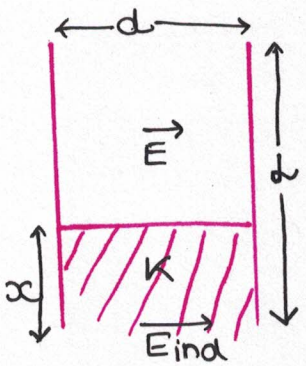
$$C_{net} = \frac{Q}{\left[\frac{Q}{A\epsilon_0}(d-t) + \frac{Q}{AK\epsilon_0} t\right]}$$

$$C_{net} = \frac{-A\epsilon_0}{d-t\left(1-\frac{1}{K}\right)}$$

If $t = d$

$$C' = \frac{KA\epsilon_0}{d} = KC$$

Case 2: We know we can solve it by thinking it in parallel and introducing an imaginary plate.



$$E = E_{ind}$$

(as potential difference is same)

$$\frac{\sigma}{\epsilon_0} = \frac{\sigma'}{K\epsilon_0}$$

$$\sigma = \frac{\sigma'}{K}$$

$$\sigma' = K\sigma$$

$$Q = (K\sigma)\frac{Ax}{d} + \left(1-\frac{x}{2}\right)A\sigma$$

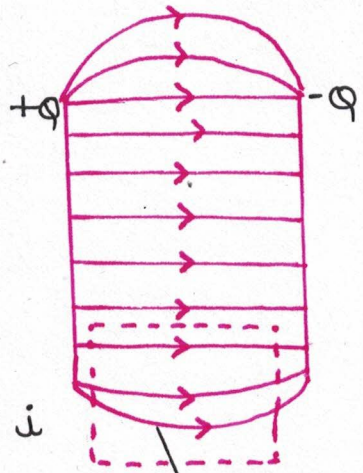
$$\Delta V = \left(\frac{\sigma}{\epsilon_0}\right)Q$$

$$C = \frac{Q}{\Delta V}$$

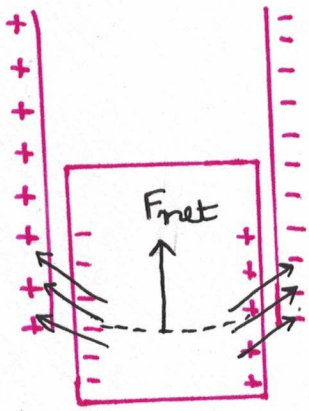
$$C = \frac{A\epsilon_0}{d\ell}\left(x(K-1)+\ell\right)$$

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FIELD LINES IN A CAPACITOR



When dielectric is inserted,



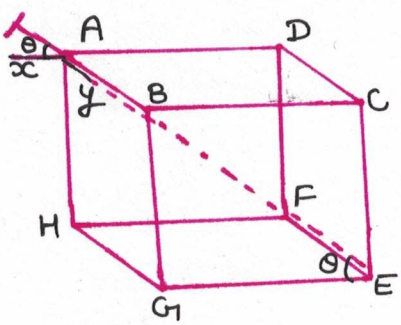
DIELECTRIC STRENGTH

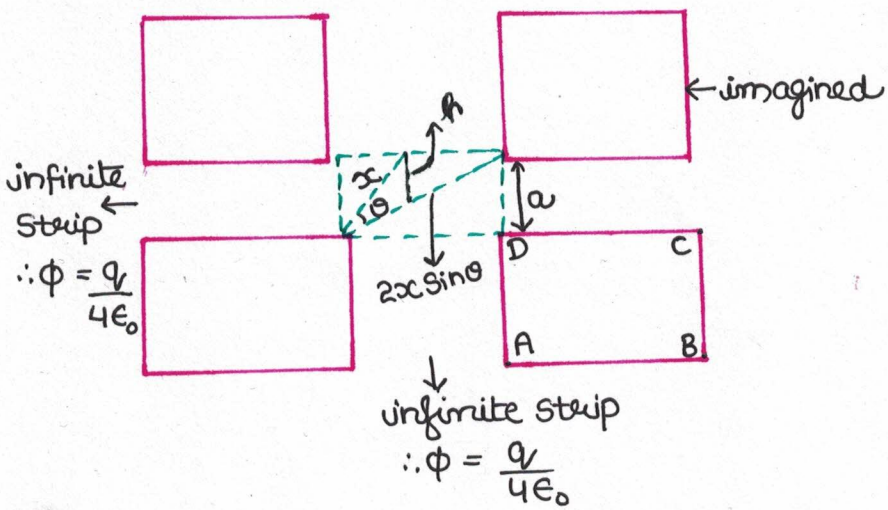
The maximum electric field that can be applied over dielectric without breakdown of material is called dielectric strength of material.

Que.) Find the flux through ABCD.

(i) $x \to 0$

(ii) $y \to 0$





$$\therefore a = \frac{2x\cos\theta}{\sqrt{2}} = \frac{2}{\sqrt{3}}x$$

$$h = x\sin\theta = \frac{x}{\sqrt{3}}$$

$$\frac{q}{4\epsilon_0} + \frac{q}{4\epsilon_0} - \frac{q}{6\epsilon_0} = \phi \text{ through empty space}$$

Total flux through the whole 4 sheets & empty space

$$= \frac{q}{2\epsilon_0}$$

$$\therefore \frac{\frac{q}{2} - \left(\frac{q}{4\epsilon_0} + \frac{q}{4\epsilon_0} - \frac{q}{6\epsilon_0}\right)}{4} = \phi_{ABCD} = \frac{q}{24\epsilon_0}$$

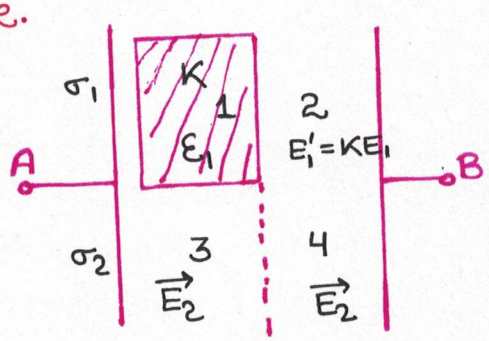
(ii) Similarly for $y \to 0$,

$$\phi_{ABCD} = \frac{7q}{24\epsilon_0}$$

(Two capacitors can be taken in parallel only if they have same potential difference.

Que.) Find the net capacitance.

1 and 3 are not in parallel as potential difference across both is not same, as electric field is different.

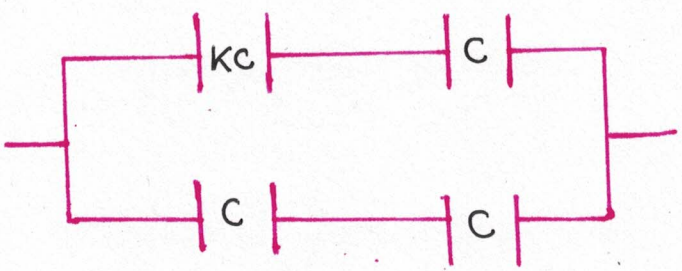


$$\left(E_1 \frac{d}{2} + KE_1 \frac{d}{2}\right) = V$$

$$E_1 = \frac{2V}{d(1+K)} = \frac{\sigma_1}{K\epsilon_0}$$

$$E_2 = \frac{V}{d} = \frac{\sigma_2}{\epsilon_0}$$

∴ Take 1 and 2 in series and then solve it with parallel to (3+4 in series)



$$C = \left(\frac{A}{2}\right)\frac{\epsilon_0}{(d/2)}$$

Or we could have done it fundamentally,

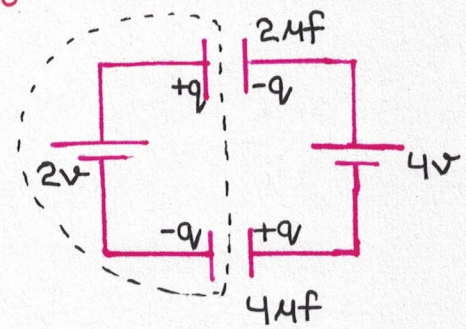
$$C_{eq.} = \frac{\sigma_1 \frac{A}{2} + \sigma_2 \frac{A}{2}}{V} \quad \left(\text{ie } \frac{\text{Total charge}}{\text{Total potential}}\right)$$

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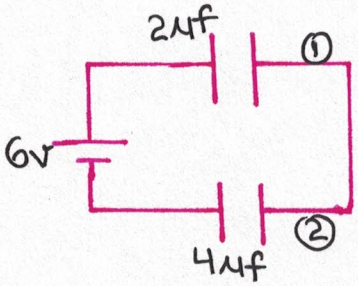
put σ_1 & σ_2 in terms of V.

Que.) Find the charges and potential of both capacitors.

we can do it by loop law.
but let's do it by an easier way



Net charge supplied by the battery is zero. It just pushes the charge.



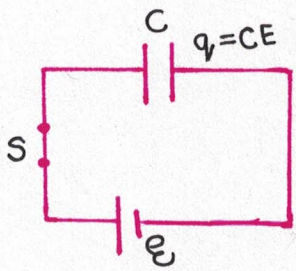
$\therefore$ Charge = 8μC

$V_1 = 4V$ and $V_2 = 2V$

HEAT DEVELOPED IN A CAPACITOR

$$W_{battery} = \Delta q \mathcal{E}$$
$$= \mathcal{E}^2 C = C\mathcal{E}^2$$

$$\boxed{W_b = C\mathcal{E}^2}$$



$$\Delta U_{cap.} = \frac{1}{2} C\mathcal{E}^2$$

$$\Delta H = \frac{1}{2} C\mathcal{E}^2$$

Here as soon as the switch is closed, the current would be flowing in the circuit with ∞ value. This creates a magnetic field in the wire. As current decreases, magnetic field decreases. Thus it creates an induced emf. The battery also works against it. Thus, the Kirchoff's law is still valid. The further detail of induced emf will be in the next chapter.

$$\mathcal{E}_{battery} - \mathcal{E}_{induced} + \frac{Q}{C} = 0 \qquad (\text{at } t=0)$$

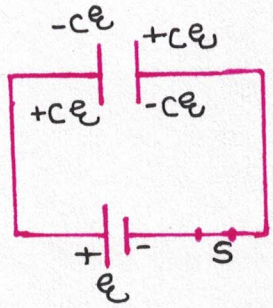
$\mathcal{E}_{induced}$ ↓ due to magnetic field

Que.) Find the heat produced?

$\Delta U_{cap} = 0$

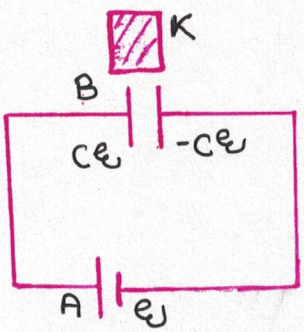
$W_{battery} = 2C\mathcal{E}^2$

$\therefore$ Heat $= 2C\mathcal{E}^2$



HEAT PRODUCED ON INSERTING DIELECTRIC (SLOWLY)

Here, we are inserting dielectric slowly. Thus, current will be finite and there will be no induced e.m.f. Thus the potential difference between the points A and B is 0. Therefore, though charge is flowing in the wire, but as there is no ΔV, heat produced in wire $= 0$.



$$\therefore W_{battery} + W_{\substack{external \\ agent}} = \Delta U_{cap.} + \Delta Heat$$

↳ we have to include the work done by external agent in inserting dielectric.

$\Delta Heat = 0$

$W_{battery} = (K-1)C\mathcal{E}^2$

$\Delta U_{cap.} = \frac{1}{2}C\mathcal{E}^2(K-1)$

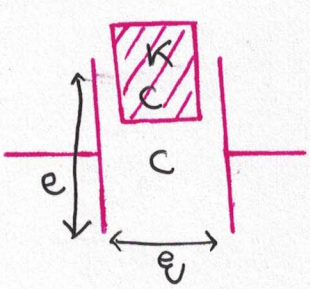
$$\therefore W_{\substack{external \\ agent}} = -\frac{1}{2}(K-1)C\mathcal{E}^2$$

FORCE ON A DIELECTRIC WHILE INSERTING DIELECTRIC (SLOWLY)

Case 1: Battery is attached

$$C_x = C\left[1 + \frac{(K-1)x}{\ell}\right]$$

$$dC_x = \left[\frac{C(K-1)}{\ell}\right]dx$$



$dW_{battery} = dC_x e^2$

$dW_{external\ agent} = -Fdx$ and $dU_{cap.} = \frac{1}{2} dC_x \cdot e^2$

$dW_{Battery} + dW_{external\ agent} = dU_{cap.}$

$\therefore Fdx = \frac{1}{2} dC_x e^2$

$F = \frac{1}{2} \frac{C(K-1)}{e} \cdot \frac{dx}{dx} E^2$

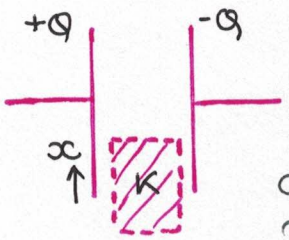
$$\boxed{F = \frac{1}{2} \frac{C(K-1)}{e} e^2}$$

Case 2: Battery is not attached

$-F.dx = dU_{cap}$

$\frac{Q^2}{2}\left(\frac{-dC_x}{C_x^2}\right) = -F.dx$

$$\boxed{F = \frac{Q^2}{2C_x}\left[\frac{C(K-1)}{e}\right]}$$



Here work done by battery = 0 as battery is not present and thus charge is constant.

CHARGE

Charge is the physical property of a particle by virtue of which it shows electromagnetic interaction.

There are two types of charges:

Conventially named as +ve and -ve

- Charges on the particles are the multiples of charge on electron
- If two or more than two bodies exchange charge among themselves, net charge on them remains constant.

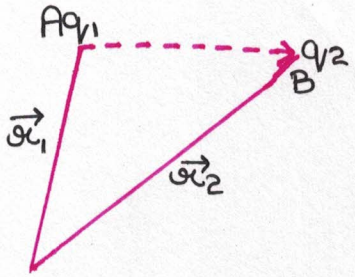
COULOMB'S LAW

Interaction b/w two charged particles is proportional to the products of their charges & inversely proportional to the square of distance b/w them.

$$\vec{F_{21}} = \frac{K q_1 \cdot q_2}{|r_2 - r_1|^3} (\vec{r_2} - \vec{r_1})$$

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$$\vec{F}_{21} = \frac{Kq_1q_2}{|\vec{r}_2 - \vec{r}_1|^2} |\hat{r}_{21}|$$

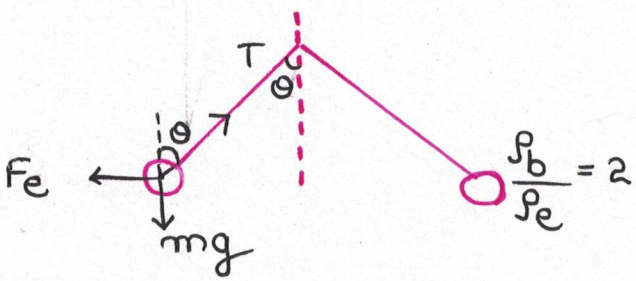


$\left(K = \frac{1}{4\pi\epsilon_0}\right)$, ϵ_0 is the permittivity of free space.

$K = 9 \times 10^9$

ELECTROSTATICS

Study of interaction b/w charge in which atleast 1 of them is at rest.



$$\tan\frac{\theta}{2} = \frac{F_e}{mg} = \frac{F_e}{W - F_B}$$

ELECTRIC FIELD INTENSITY

Intensity of electric field is defined as force per unit charge.

$$\vec{E} = \lim_{q \to 0} \frac{\vec{F}}{q}$$

$$\vec{F} = q\vec{E}$$

$$\vec{E} = K\left(\frac{Q}{r^2}\right)\hat{r}$$

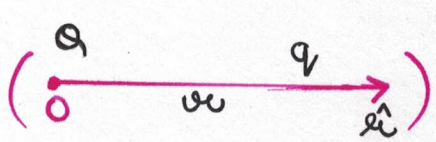
$$\vec{E} = \sum \vec{E_i}$$

FIED DUE TO POINT CHARGE

$$\vec{E} = \frac{K(Q)}{r^2} \cdot \hat{r}$$

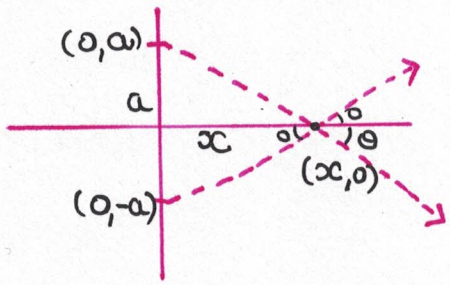
$$\vec{E}_{net} = \sum \vec{E_i}$$

or $\vec{E}_x = \int dE_x$



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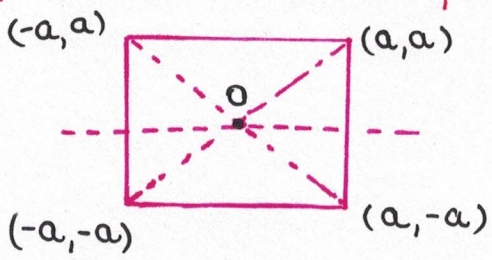
Que.) Two point charges each of magnitude 'q' are placed at Q (0,a) and Q (0,-a). Find electric field at P(x,0)



$$\vec{E} = 2F\cos\theta$$

$$= \frac{2KQ}{(a^2+x^2)} \times \frac{x}{(a^2+x^2)^{1/2}}$$

Que.) Four point charges each of magnitude Q are placed at points of square. Find electric field.



$$\vec{E} = \frac{4KQ \times x}{(r^2+x^2)^{3/2}}$$

$$r = \frac{a}{\sqrt{2}} \quad \& \quad x = 2$$

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